

Reconstructing Reliability: Predicting and Tracing Delay on SEPTA Regional Rail

Katie Steele

[GitHub repository](#)

1 Introduction

The SEPTA Regional Rail network is one of the largest commuter rail systems in the United States, serving more than 150 stations throughout the Philadelphia region and carrying an average of 84,700 weekday riders.¹ Despite its scale, the system continues to struggle with constrained funding and inconsistent reliability. SEPTA’s public reporting on system reliability centers on its on-time performance (OTP) rate, which defines a train as on time if it reaches its terminus within six minutes of its scheduled arrival. SEPTA’s annual system-wide Regional Rail OTP target is 90 percent,² but the agency has not met that benchmark in any year dating back to at least 2017 (N8).

Annual OTP rate provides a useful summary of overall performance, but it does not explain how delays develop or where they concentrate. To better understand what’s driving underperformance in the aggregate, this analysis asks two related questions:

1. Before departure, how much of a train’s eventual delay risk can be anticipated from information already available?
2. During the run, where and under what conditions do delays concentrate?

Both questions start from the same public record — SEPTA’s live delay feed — but reconstruct it differently: the first collapses each run down to a single terminal outcome, while the second preserves a run’s full stop-by-stop trajectory.

The results show that Regional Rail delay has identifiable structure in both cases. Before departure, recent train, line, and network history provide substantial predictive signal for delay risk — contributing more to model performance than contemporaneous context like weather, holidays, sports schedules, or rush-hour timing — but the signal is uneven at the line level and remains incomplete. During the run, delay accumulation is similarly uneven, with the worst performance concentrating on outbound, longer-distance, and Amtrak-corridor service.

More broadly, this analysis illustrates both the strengths and the limits of public operational data. The records used here are sufficient to recover meaningful predictive signal before departure and identify persistent spatial delay patterns during the run, but they cannot fully explain why certain delays emerge or why specific corridors consistently underperform. The sections that follow identify the boundary between what can be inferred from publicly available data and what requires richer information.

2 Core Data

The analysis is built on a public record of roughly 25 million lateness observations collected from SEPTA’s live delay feed ([archived by William Entriken](#)), polled by the minute since 2011. To keep coverage consistent, the record is restricted to roughly 20 million observations spanning 2017–2025. The feed is designed for real-time passenger use, not historical analysis; each observation captures only a timestamp, a train “block” identifier, and a reported lateness value in minutes — not a structured record of the trip it belongs to (N1).

¹ “SEPTA Regional Rail,” *Wikipedia*, https://en.wikipedia.org/wiki/SEPTA_Regional_Rail.

² SEPTA, *Regional Rail Reliability Improvement Plan*, <https://www.septa.org/operations/service-performance/regional-rail/>.

On its own, a snapshot like this is nearly unusable: it carries no line, direction, or schedule, so there’s no way to reliably map it to any particular service. SEPTA’s [General Transit Feed Specification \(GTFS\)](#) releases close that gap — they provide historical schedule data identifying each run’s line, direction, origin, terminus, and scheduled timing across the full nine-year study period. Matching every delay observation against the GTFS release active on its date turns anonymous delay pings into identifiable train runs, the foundation both analyses that follow are built on (N2b).

3 Before Departure: How Predictable is Delay?

This section asks how much of a train’s eventual delay risk can be anticipated using only information available before departure. The goal is to estimate elevated risk, not to prove why a delay will occur.

3.1 Data and run-level reconstruction

For my pre-departure analysis, I reconstruct the ~20 million delay feed snapshots into 1,551,754 individual train runs — the primary unit of analysis for predictive modeling. Each run’s final delay snapshot is treated as its terminal delay, and the predictive model’s outcome variable follows SEPTA’s OTP definition: `is_otp` equals 1 when a run’s terminal delay is less than 6 minutes, 0 otherwise (N7).

3.2 Supplementary data

Matching SEPTA’s delay feed and GTFS gives line, direction, and schedule, but nothing about demand or same-day conditions. To fill that in, the predictive modeling pipeline layers in additional [ridership data](#) (N4/N4b), holiday and major league sports schedules (N5), and hourly weather data (collected, cleaned, and compiled by Charlotte Mansfield in `weather/`; prepped in N6) (Table 1). This reconstruction transforms the passenger-facing delay feed into a longitudinal dataset suitable for predictive analysis at the individual train-run level.

Data source	Role in analysis	Ref
SEPTA live delay feed	Reported delay over time; terminal lateness; OTP outcome	N1
SEPTA GTFS releases (~333 historical releases)	Run identification; line, direction, origin, terminus, scheduled timing	N2b
Ridership data	Line-level demand context	N4/N4b
Holiday, service-exception, sports calendars	Same-day contextual features	N5
NOAA hourly weather	Weather conditions available by scheduled departure	N6

Table 1: Predictive dataset — data architecture at a glance

3.3 Modeling design and evaluation

Every feature used for prediction is anchored to a run’s scheduled departure time and reflects only information that would have been observable at or before that moment. To prevent temporal leakage, outcome features (`lateness` and `is_otp`) are dropped from the feature set and all weather and operational history features are lagged.

This section’s baseline model comparison below uses a two-way split, training on 2017–2024 and evaluating on 2025, with no validation set. Sections 3.4 onward use a three-way split: training on 2017–2023 ($n = 1,170,530$), with 2024 ($n = 187,033$) held out as a validation set and 2025 ($n = 126,332$) retained as an untouched test set throughout. As the worst-performing year in the entire dataset (76.23% OTP across the full calendar year; Table 5), 2025 serves as a rigorous test.

Because the 2025 test set is imbalanced, with 78.6% true on-time runs, a naive classifier that always predicts “on time” would achieve 78.6% accuracy while identifying none of the delayed runs. Accuracy therefore provides a misleading measure of performance in this setting, so this analysis instead emphasizes ROC-AUC and late-class precision-recall AUC (PR-AUC), better metrics under class imbalance.

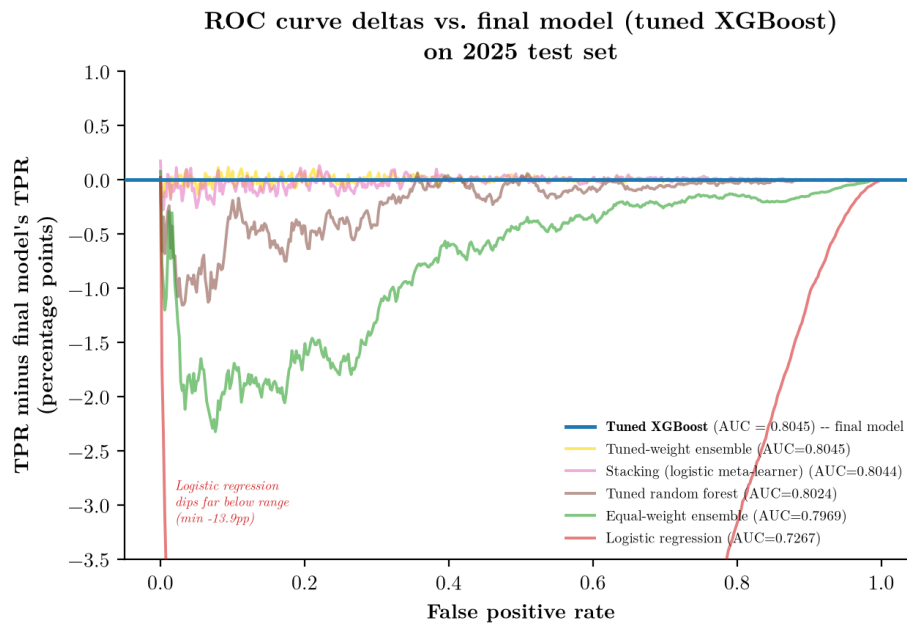
Initial benchmarks showed that tree-based methods captured available signal more effectively than a logistic regression with line fixed effects: the logistic model reached 0.732 ROC-AUC and 0.455 PR-AUC, while an XGBoost classifier and random forest reached ROC-AUC / PR-AUC of 0.807 / 0.614 and 0.793 / 0.589 respectively (Table 6). This was an early indication that non-linear interactions in the data provided more signal than individual linear relationships.

3.4 Stacked prediction feature

Each of the initial models incorporated a large set of features I’d engineered from the supplementary datasets, but the biggest performance gain came from a new feature that distilled their signals into a single model-generated estimate. After training a second gradient-boosting model to predict a run’s continuous $\log_{1p}(\text{lateness})$, I fed that prediction (`pred.lateness_gbm`) back into the binary XGBoost classifier as a stacked lateness feature. On the held-out test set, adding this single feature improved XGBoost’s ROC-AUC by 0.5 points and late-class PR-AUC by 0.8 points (N9b).

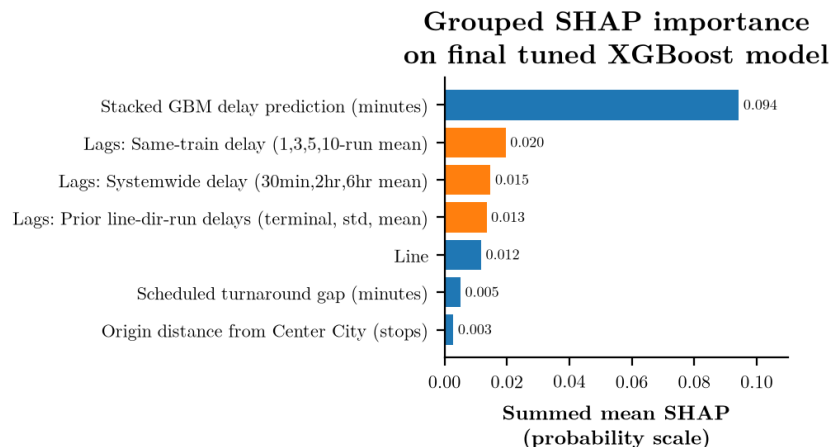
3.5 Ensembling and final model

I next tested several ensemble approaches for the binary classification task, combining logistic regression, random forest, and gradient boosting through equal weights, tuned weights, and stacking, but none meaningfully outperformed the tuned XGBoost classifier with the stacked lateness feature (N9c, N9d). The simpler model was therefore retained for interpretability and reproducibility.



On the held-out 2025 test set, the final model reaches 0.8045 ROC-AUC and 0.610 late-class PR-AUC. In practice, how many delays the model flags depends heavily on the decision threshold chosen: shifting from the default 0.5 threshold to a more sensitive one doubles the share of delays caught (36.6% to 73.2%) but more than triples the false-alarm rate (11.1 to 37.9 per 100 runs) (N9e).

3.6 Feature importance



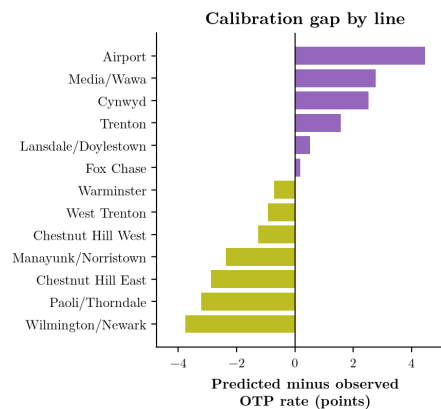
A SHAP-based analysis, grouping correlated features to measure their combined contribution (N10, §1), shows `pred_lateness_gbm` alone contributes 5 times more to the model’s prediction than any other feature group. Still, recent history at the train, line, and network level carries independent weight alongside the stacked lateness feature.

Permutation importance on `pred_lateness_gbm` itself (N10, §2) tells us what predicts the predictor: Its top 10 features comprise almost the exact same set as the classification model — operational history lags, plus line and scheduling pressure. In other words, the classification model’s strongest feature draws very little predictive signal from context outside of the Regional Rail system; it’s essentially a measure of operational momentum. A robustness check on the final binary classifier tells a consistent story: cutting the 72-feature set down to its 40 most informative features barely moves any performance metric (N10, §4), meaning the excluded features weren’t carrying useful signal to begin with.

The next question is which delays the model cannot see coming.

3.7 Model errors: Where advance warning breaks down

Overall, the model is well calibrated at the network level — for example, runs in the lowest-risk decile are predicted to be on time 97.6% of the time, compared with 97.0% observed — but aggregate performance masks meaningful variation across lines. The model is overly optimistic on the Airport Line and underconfident on Paoli/Thorndale and Wilmington/Newark, the two weakest-performing lines in the network (Section 4.1). Even within individual lines, the errors are unevenly distributed: on the Airport Line, the largest gaps are concentrated among midday inbound trains and especially pronounced on Thursdays (N9e). These initial findings suggest that any practical deployment of this model would require a comprehensive calibration analysis.



More fundamentally, some delayed runs do not carry a detectable warning in the data at all. Among actually late trains in a held-out sample, correctly flagged runs carry a strong signal from the stacked lateness feature (mean SHAP contribution -0.312) and the train’s own most recent run (-0.014), whereas missed runs show almost no contribution from either (-0.056 and $+0.001$ respectively — N10, §3). These false negatives appear difficult to anticipate from the available pre-departure data; the delays could reflect an event occurring after departure, feed noise, or a limitation of the model itself, but the data here are not sufficient to explain them. Any use of model outputs should therefore be framed as a risk estimate and not as a guarantee that every delay will be detected.

The next section shifts from examining individual train runs to examining systematic differences across lines, directions, distance, and track environments.

4 During the Run: Where Do Delays Concentrate?

This section presents descriptive and regression-based findings across all available years. It asks whether unreliable service is evenly distributed or persistently associated with particular parts of the network, and identifies operating patterns linked to underperformance.

4.1 Line and directional variation

Descriptive analysis shows that performance varies substantially by line: Paoli/Thorndale posts the lowest OTP at roughly 75.8%, while Cynwyd averages 92.6% across the nine examined years. Direction matters almost as much as line: Across the network, outbound runs leaving Center City finish more than a full minute later on average than inbound runs (+1.054 minutes, $n = 751,412$ outbound / 800,342 inbound) — the largest gap associated with any binary variable in the dataset.

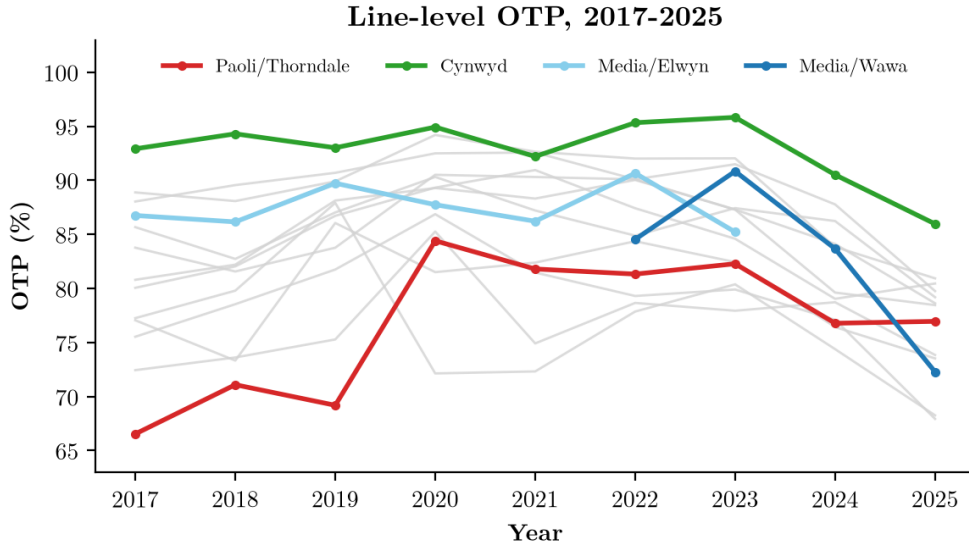


Figure 1: OTP by line, 2017–2025 — most lines dip in 2024–2025. Highlighted lines Cynwyd and Paoli/Thorndale represent the best and worst average OTP across dataset. Media/Elwyn and Media/Wawa also highlighted to showcase the 2022 line extension; they briefly coexisted in the delay feed data.

This directional gap raises a natural next question: where along a route does it actually emerge? Investigating it requires a finer unit of analysis than run-level totals; the next section reconstructs lateness at the individual stop level.

4.2 Data and stop-level reconstruction

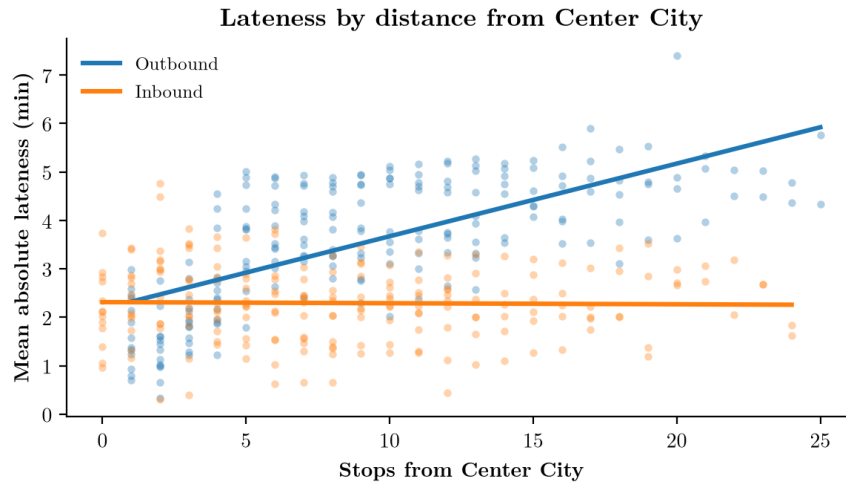
For this analysis, I return to Section 2’s roughly 20 million-row delay feed. Section 3 reduced those observations to one terminal outcome per run; here, I instead reconstruct each run’s lateness at every scheduled stop along its route. Because the archived delay feed records a new observation only when a train’s reported lateness changes, I use Last Observation Carried Forward to recover the value in effect at each scheduled stop (N11, §3; N2c), yielding 22.8 million stop visits. From there, I derive two quantities used throughout the section: mean lateness by line, stop, and direction; and, for each stop-to-stop segment of track (a “hop”), how much delay a run accumulates or recovers while crossing it. Each of the 20.2 million hops is further tagged by whether it falls on SEPTA- or Amtrak-owned track (N11, §2). Unlike the predictive analysis, this section uses the full 2017–2025 history with no train/validation/test split, since the goal is to characterize where delay concentrates system-wide, not to hold out data for evaluation.

Data source	Role in analysis	Ref
SEPTA live delay feed (same feed as Table 1)	Stop-level lateness via LOCF reconstruction	N1, N11
SEPTA GTFS stop times + cross-release stop-ID crosswalk	Scheduled stop sequence; consistent stop identity across renumbering	N2c
Infrastructure / track ownership	Amtrak-versus-SEPTA corridor tagging	N11

Table 2: Spatial dataset — data architecture at a glance

4.3 Distance x directional accumulation

Lateness is also associated with distance from Center City, and the association is direction-specific (N11, §4). Evaluating individual hops of track across the Regional Rail system, I find that each additional hop away from Center City is associated with roughly 5.9 extra seconds of mean absolute lateness on outbound service, holding Amtrak ownership and line fixed effects constant ($p < 0.001$). Inbound service shows a smaller opposite effect: about 2.5 fewer seconds of lateness per hop ($p < 0.001$). If these effects compound linearly, this delay accumulation could be substantial. On the Wilmington/Newark line, which has 22 stops, the findings would suggest an added outbound delay accumulation of almost 2.2 minutes stemming from this interaction alone. Inbound, the same run would make up about a minute of lost time.



This interaction is interesting in that it separates the geographic effect, which we might expect, from an asymmetry between inbound and outbound service. It's easy to imagine a train racking up small delays at each stop on a long line, due to dis/embarications or other frictions — but we only see worsening per-hop delays on longer outbound service; comparable inbound travel manages to gain time. This could be due to any number of differences between the two services (scheduling, infrastructure quality, dispatch procedures) and would benefit from further investigation.

4.4 Amtrak track-sharing

About half of SEPTA's Regional Rail lines rely partially on Amtrak-owned service corridors and therefore share track with Amtrak trains. To test whether shared infrastructure is associated with delay accumulation, I run a weighted least-squares regression of hop-level lateness on Amtrak-ownership status, controlling for distance from Center City, line, and direction. It shows that Amtrak-owned segments run about 1.09 minutes later outbound ($p = 0.003$), and about 0.49 minutes later inbound ($p = 0.059$) than SEPTA-owned segments on average.

To further investigate exactly where along each trip this extra time accumulates, I tag every track hop

across the dataset into one of four categories according to its ownership: all SEPTA track, all Amtrak track, a SEPTA-to-Amtrak entrance transition, or an Amtrak-to-SEPTA exit transition. Then, for each type, I measure the average delay gained or recovered specifically crossing that one segment.

Segment type	Avg. change in lateness	Segments (n)
Entering Amtrak-owned track	+0.85 min	529,825
Exiting Amtrak-owned track	+0.74 min	421,106
Fully inside Amtrak-owned track	+0.12 min	5,925,611
Fully inside SEPTA-owned track	+0.15 min	12,886,549

Table 3: Time lost per segment, by segment type (full 2017–2025 history)

The result is clear: Trains tend to accumulate 5 to 7 times as much delay on the ownership boundary segments compared to the segments that lie on either side of them. The pattern holds up consistently across the 6 lines that share Amtrak corridors (N11, §4b); SEPTA appears to deal with substantial handoff friction right at the Amtrak boundary line. Track ownership is likely to act as a proxy for several corridor characteristics (congestion, scheduling, dispatch priority, merging, queuing) that are not observed in this data, but this finding warrants further exploration.

5 Beyond Arrival: Implications and Future Work

The predictive and spatial analyses reach different limits with the available public data. Section 3 recovers substantial pre-departure signal but leaves some delays without detectable warning, while Section 4 identifies persistent geographic and operating relationships without being able to say with certainty what relationships or factors produce them. Both limitations point toward richer internal operational data as the most useful next step. The following three recommendations translate these findings into targeted areas for future work.

5.1 Pilot a pre-departure predictive risk signal

In 2025, roughly one in four Regional Rail runs were late. The predictive results suggest that this unreliability is more chronic than episodic: the model’s strongest signal comes from recent operational history at the train, line, and network levels, while contemporaneous factors like weather, calendar effects, sports schedules, and rush-hour timing contribute substantially less. A pre-departure risk model built with SEPTA’s richer internal dispatch, service alert, infrastructure, and mechanical data could test how much additional predictive power exists beyond the baseline developed here on a patchwork of public datasets.

The value of this modeling would depend on its intended use. As an internal tool for dispatchers, it could provide early detection of trains facing elevated delay risk and help prioritize attention or resources accordingly. Threshold selection could be tuned to operational goals (for example, favoring broader detection or fewer false alarms) and refined over time as additional data and feedback become available.

Before deployment, two findings from Section 3.7 would require much closer attention: calibration would need a more comprehensive evaluation, and false negatives would require deeper investigation, since the average false negative uncovered in this analysis carried almost no detectable pre-departure signal at all.

5.2 Audit the distance x direction interaction

Trains traveling away from Center City tend to accumulate delay as they move farther along the line, while inbound trains recover time. A general distance penalty would point at geographic effects — maybe infrastructure conditions worsen farther from the city — but the directional asymmetry suggests that differences in scheduling (recovery time) or operations (dispatch procedures) may also contribute. This gives SEPTA a specific operational question to investigate: What differs between inbound and outbound service for trains from Center City?

On the Wilmington/Newark Line, the average outbound distance-by-direction interaction alone could push a run roughly one third of the way past the entire six-minute OTP threshold. More broadly, the five lowest-performing lines represent some of SEPTA’s longest routes and account for roughly 42% of all runs (Table 9). System-wide, 3.3% of observed runs finish just beyond the OTP threshold, between six and seven minutes late (N8) — meaning that even modest one-minute reductions in accumulated delay could materially affect the annual topline. An audit of directional discrepancies on these lines could reveal whether scheduling, infrastructure, dispatch, or other operating practices offer opportunities to reduce that gap.

5.3 Investigate the Amtrak track-sharing penalty

On average, entering Amtrak-owned track costs trains 0.85 minutes and exiting costs them 0.74, while delays on segments entirely inside Amtrak territory (+0.12) behave essentially like segments entirely inside SEPTA territory (+0.15). If full Amtrak-owned corridors were more congested or in worse repair, we would expect delay accumulation to persist across them in full rather than concentrate at the transition points. Instead, the pattern suggests a per-crossing cost associated with ownership boundaries, potentially due to merge points, priority decisions, queuing, speed restrictions, or dispatch coordination.

According to these findings, a run entering and exiting an Amtrak corridor accumulates roughly 1.3 minutes of delay beyond what it would lose on equivalent SEPTA-only track — almost a quarter of its maximum OTP allowance. Explaining why requires records beyond the scope of this dataset, like track assignments, signaling logs, and restrictions at the transition points, but the finding narrows a diffuse problem with shared infrastructure into a specific diagnostic question about efficiency at track ownership boundaries.

6 Conclusion

This project began with a roughly 25 million-row archive of passenger-facing delay observations containing only a train “block” identifier, timestamp, and reported lateness. By integrating schedule, ridership, calendar, weather, and infrastructure data, I reconstructed that archive in two different ways: into 1.5 million individual train runs for predicting delay risk before departure, and into a 22.8 million-row stop-level table for tracing how delays develop during the run. Together, these datasets recovered meaningful patterns at two different scales: when a run is likely to be late, and where delays tend to accumulate once it’s underway.

However, the remaining uncertainty in this analysis shows where public data begin to fall short. Even after engineering an extensive set of features describing conditions outside the Regional Rail system, the classification model’s strongest predictive signal by far came from the system’s own recent operational history — recent train, line, and network conditions consistently outweighed other contextual factors like weather or calendar effects. Model performance quickly plateaued, and cutting the feature set nearly in half produced almost no change in performance. Together, these results suggest that the available data had little more to give.

The stop-level analysis revealed persistent spatial structure, with unreliable service concentrated along particular corridors and operating settings. But its ceiling was explanatory: It could identify problem routes and delay-prone operating patterns but could not support stronger conclusions about what produces them. In all, the findings suggest that the biggest remaining gains would depend less on additional modeling complexity or external data integration than on access to deeper operational information.

My original datasets are sufficient to identify where delay risk has momentum, where unreliable service concentrates, and which relationships warrant deeper investigation. Connecting those patterns to dispatch, equipment, infrastructure, crew, and other internal records would be the next step toward moving from prediction and description to diagnosis.

7 Limitations

7.1 Predictive limitations

Cascading delays. SEPTA points to cascading (knock-on) delays as a major driver of system underperformance,³ and I can’t really test that claim here. Every line funnels through the Center City trunk — by far the busiest, hardest-to-manage part of the system, and the spatial data do show delays clustering there — but each trunk station has 4+ tracks that this dataset can’t tell apart. Without knowing which physical track a train actually used, I can’t confirm two trains shared a track closely enough to call one the cause of the other’s delay. Single-track lines like Cynwyd would make a cleaner test case, but they run much less frequently (often about once an hour), which is a poor setting to analyze the relationship between headway and knock-on-effects. I tried a couple versions of this analysis and dropped it for these reasons. That said, how strongly recent line- (and to a lesser extent, network-) level history predicts delay is a pretty good hint that knock-on delay is a big piece of the puzzle.

Model drift. I only validated the model against one future year, and weather data ran out part of the way through it (coverage stops August 26, 2025). A quick check against the validation set suggested the model was pretty robust seasonally — so not having winter in the test set was not an obvious negative impact — but this deserves checking against other holdout years and monitoring for drift over time.

Real-time service alerts. I couldn’t find any archive of SEPTA’s service alerts, which are the disruption and short-notice changes that get pushed to riders in real time. The only “exceptions” data that exists publicly (via GTFS) is sparse and only covers changes planned well in advance. My guess is this is one of the biggest gaps in the predictive model; real-time alerts might make a huge difference.

COVID-19. Table 5 shows 2020 as the best OTP year on record (87.2%) and 2021 close behind — almost certainly reflecting collapsed ridership/lighter schedules and not a genuinely more reliable system. Ridership is the strongest single correlate of lateness I found, so pooling those two pandemic years in with everything else could be distorting what the model treats as “normal” operational history, especially in the lagged features.

7.2 Spatial and structural limitations

Socioeconomic delay exposure. I also attempted a Census-based demographic analysis which didn’t make it into my full pipeline (or this report). My approach was centered around station-area Census measures, but they are likely poor proxies for actual riders, especially as a line moves through shifting geographies. A half-mile buffer seems very unlikely to hold as we move toward the outer edges of the network, e.g., to Doylestown Station, where you’d expect to see more park-and-riders commuting to the station from far outside a half-mile radius. The Airport Line altogether is an outlier; it likely has very different ridership patterns than other lines, and runs through an unusual corridor. Some spatial corridor analysis did suggest that tracks move through lower-income areas, but this could very easily be reverse causality; findings overall were mixed and mostly statistically insignificant using my naive approach.

Amtrak boundaries. I built the Amtrak/SEPTA ownership boundaries from a relatively quick search, not an official dataset,⁴ so the exact stop where ownership flips could be off by a station or two in places. This is worth a caveat since a misplaced boundary would really change the headline finding for that analysis.

³ SEPTA, *Regional Rail Reliability Improvement Plan*, <https://www.septa.org/operations/service-performance/regional-rail/>.

See also Andrew Busch (SEPTA spokesperson), quoted in “SEPTA Regional Rail delays this morning are due to a train pulling down overhead wires,” *The Philadelphia Inquirer*, January 20, 2026, <https://www.inquirer.com/transportation/septa-regional-rail-delays-downed-wires-weather-20260120.html>.

⁴ For a public secondary source on the SEPTA/Amtrak/CSX/City of Philadelphia split, see “SEPTA Regional Rail,” *Wikipedia*, https://en.wikipedia.org/wiki/SEPTA_Regional_Rail.

Acknowledgements

Many thanks to my advisor, Sam Sengenito, and Charlotte Mansfield, for introducing me to the delay feed data and for their constructive feedback throughout the summer. I am especially grateful to Charlotte for navigating the many complexities of NOAA data to develop the weather dataset used for the predictive model.

AI Usage

I used Claude Code for assistance with data pipeline and project maintenance toward the end of the summer, including formatting markdown and section summaries, general debugging that stemmed from pipeline restructure, and building fetch, plotting, and caching helper functions to standardize across the project. For submission, I used it to create my appendix, format citations/references, and manage GitHub commits.

Notebook References

Every notebook cited in the main text, in pipeline run order. Ref is the short key used for in-text citations (e.g. N7); section locators (§1, §2, ...) are given only for N10 and N11, which each back several distinct findings cited in different places.

Ref	Notebook (.ipynb)	Description
N1	01_otp_load	Loads and cleans the raw SEPTA delay feed.
N2b	02b_gtfs_merge	Resolves each delay observation to a line, trip, direction, and scheduled time via GTFS matching; produces the run-identifiable dataset both analyses build on.
N2c	02c_gtfs_stops_crosswalk	Cross-release stop-ID crosswalk, resolving SEPTA’s stop renumbering (most substantially in 2022) so historical runs map onto a consistent set of stop identities.
N4/N4b	04_ridership / 04b_ridership_safe	Line-level ridership data, cleaned and leak-safe for use as a predictive feature.
N5	05_calendar_events	Holiday, service-exception, and major league sports schedules.
N6	06_weather_prep	Hourly NOAA weather, prepared as of scheduled departure time.
N7	07_reduce_for_modeling	Builds the run-level predictive dataset (1,551,754 runs) and its features, including feature-timing safety (GBM.FEATURES, defined in <code>utils.py</code>).
N8	08_descriptive	Descriptive OTP analysis: systemwide by year (Table 5), by line, and threshold sensitivity.
N9	09_prediction	Baseline model comparison: logistic regression, XGBoost, random forest (Table 6).
N9b	09b_stacked_feature	Builds and tests the stacked lateness-regression feature.
N9c	09c_tuning	Hyperparameter search for the final model.
N9d	09d_ensemble	Tests ensemble approaches against the tuned model alone (Table 7).
N9e	09e_error_analysis	Threshold/confusion-matrix analysis and per-line calibration.
N10	10_feature_importance	Four-section feature-importance analysis: §1 grouped SHAP ranking, §2 regression-GBM permutation importance (Table 8), §3 late-class SHAP (caught vs. missed), §4 reduced-feature robustness check.
N11	11_spatial_delay_patterns	Spatial analysis across the full 2017–2025 GTFS history: §2 Amtrak/SEPTA track-ownership tagging, §3 row-level LOCF exposure reconstruction, §4 distance/Amtrak-ownership regression, §4b ownership-boundary-crossing analysis (Table 3).

Table 4: Notebook references, in pipeline run order

Appendix: Supplementary Tables

Year	2017	2018	2019	2020	2021	2022	2023	2024	2025
OTP	79.98%	80.70%	84.31%	87.19%	85.38%	85.77%	85.66%	80.80%	76.23%

Table 5: Systemwide OTP by year (full run-level table, lateness < 6)

Model	ROC-AUC	PR-AUC (late)	Accuracy
Naive (base rate)	0.500	0.214	0.786
Logistic regression + line FE	0.732	0.455	0.800
XGBoost	0.807	0.614	0.832
Random forest	0.793	0.589	0.829

Table 6: Baseline models (2025 test set, pre-stacking feature set, default hyperparameters, threshold = 0.5)

Model	ROC-AUC	PR-AUC (late)	Recall (late)	Precision (late)	Bal. acc.	Accuracy
Logistic regression	0.727	0.450	0.186	0.594	0.576	0.799
Tuned XGBoost (final)	0.8045	0.610	0.366	0.706	0.662	0.832
Tuned random forest	0.802	0.608	0.361	0.707	0.660	0.832
Equal-weight ensemble	0.797	0.594	0.325	0.738	0.647	0.831
Tuned-weight ensemble	0.8045	0.610	0.366	0.706	0.662	0.832
Stacking (meta-learner)	0.8044	0.610	0.388	0.684	0.670	0.831

Table 7: Final model comparison (2025 test set, full feature set including the stacked feature)

Rank	Feature	R ² drop when shuffled
1	lag_train.lateness_1run_mean	0.0986
2	lag_train.lateness_10run_mean	0.0637
3	lag_line.lateness_min	0.0475
4	sched_turnaround_gap_min	0.0468
5	lag_network.lateness_2hr_mean	0.0430
6	lag_network.lateness_30min_mean	0.0286
7	sched_duration_sec	0.0230
8	lag_train.lateness_3run_mean	0.0217
9	line	0.0199
10	sched_origin_sec	0.0119

Table 8: Regression GBM permutation importance (source of pred_lateness_gbm; 2025 test set, loglp(lateness) space)

Line	OTP	95% CI	Mean lat. (min)	Median lat. (min)	n runs
Paoli/Thorndale	75.8%	[75.6, 76.0]	4.34	1.0	169,405
Wilmington/Newark	76.2%	[76.0, 76.5]	4.57	2.0	105,373
West Trenton	77.0%	[76.8, 77.2]	4.08	1.0	111,752
Lansdale/Doylestown	78.0%	[77.8, 78.2]	4.19	2.0	138,198
Trenton	79.6%	[79.4, 79.8]	4.01	1.0	130,297
Media/Wawa	82.0%	[81.6, 82.3]	3.63	1.0	49,555
Warminster	84.0%	[83.8, 84.2]	3.31	2.0	152,723
Chestnut Hill East	85.2%	[85.0, 85.4]	2.93	1.0	93,322
Manayunk/Norristown	85.3%	[85.1, 85.5]	2.92	1.0	120,684
Chestnut Hill West	85.7%	[85.5, 85.9]	2.73	0.0	87,197
Fox Chase	87.3%	[87.1, 87.5]	2.75	1.0	98,185
Media/Elwyn (pre-2023)	87.9%	[87.6, 88.1]	2.63	0.0	69,478
Airport	88.8%	[88.6, 88.9]	2.42	1.0	199,663
Cynwyd	92.6%	[92.3, 93.0]	1.77	0.0	25,922

Table 9: Line-level OTP, full 2017–2025 history (95% CI on the proportion)